

arXiv: 2604.08377

SkillClaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Agent skills continuously evolve by aggregating cross-user trajectories and applying agentic, evidence-driven updates.

Multi-user interaction

Session collection

Skill evolution

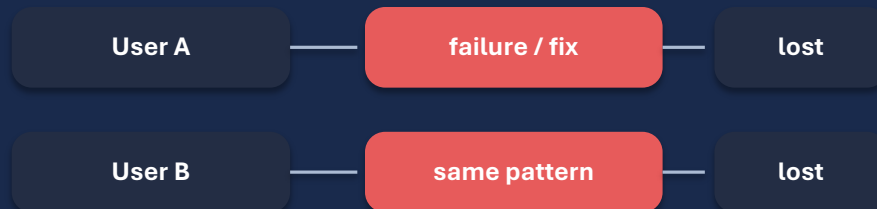
Skill synchronization

Agent Skills Remain Static After Deployment

Users repeatedly rediscover failures and recoveries that never accumulate at system level

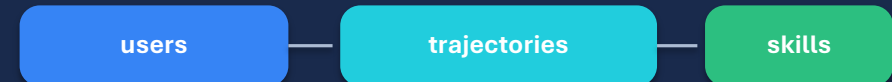
Today: static skill paradigm

Skills are manually installed from a centralized hub. Runtime discoveries rarely persist beyond individual sessions; similar tasks recur, but the system does not improve.



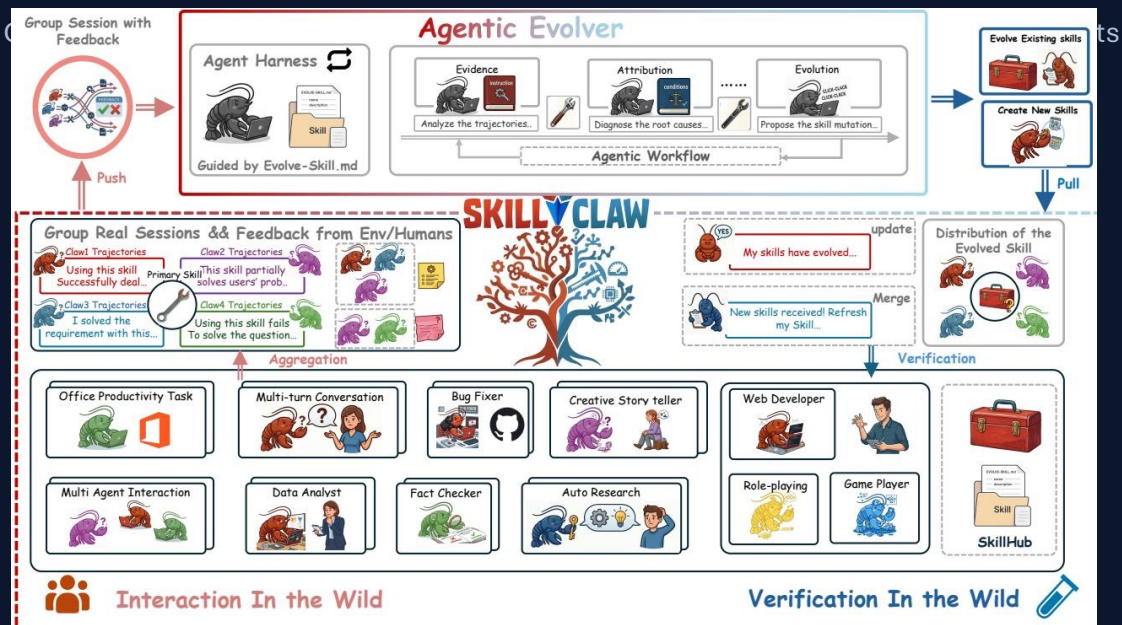
SkillClaw: evolving skill paradigm

Cross-user trajectories become a shared evidence base. Verified skill mutations are merged and synchronized back to all agents.

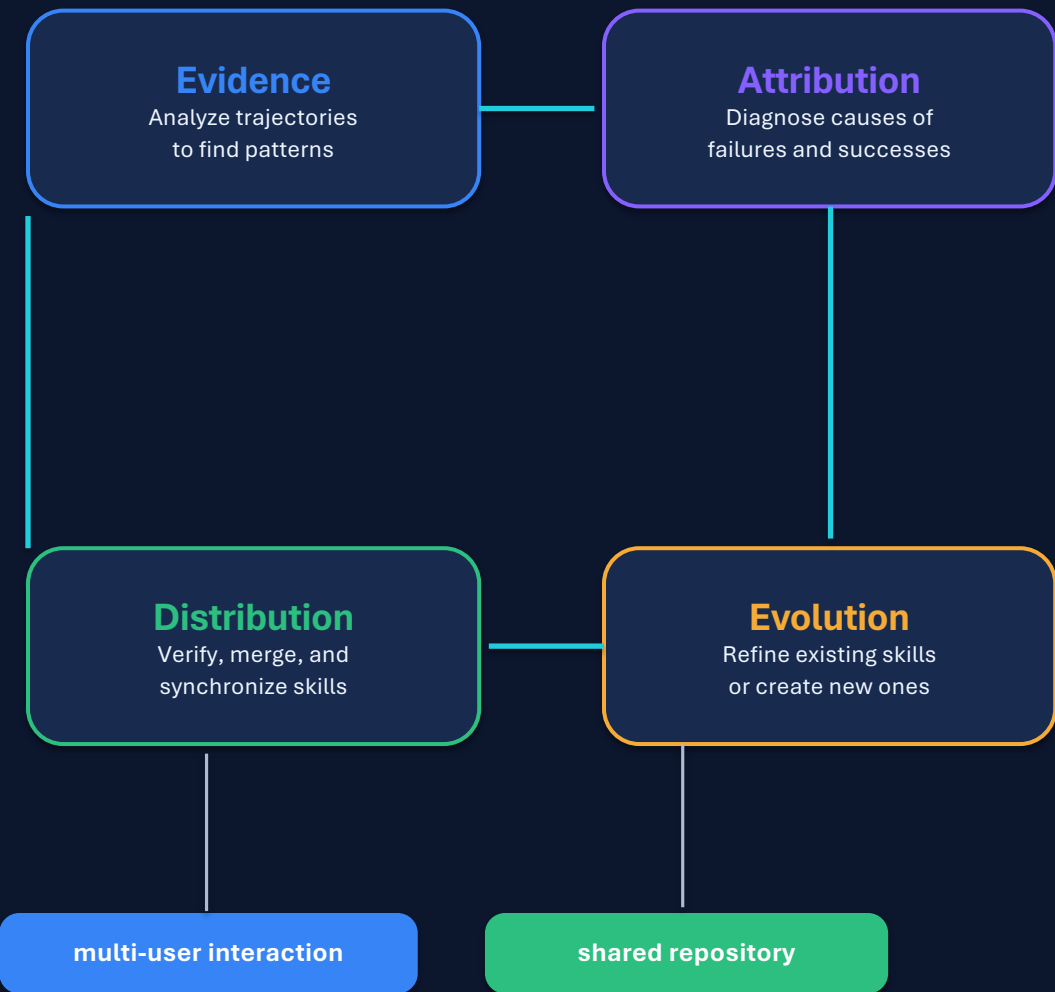


Memory methods remain instance-tied; skill methods compress experience but treat libraries as static.

Closed-Loop Pipeline: Interaction Becomes Shared Skill Evolution

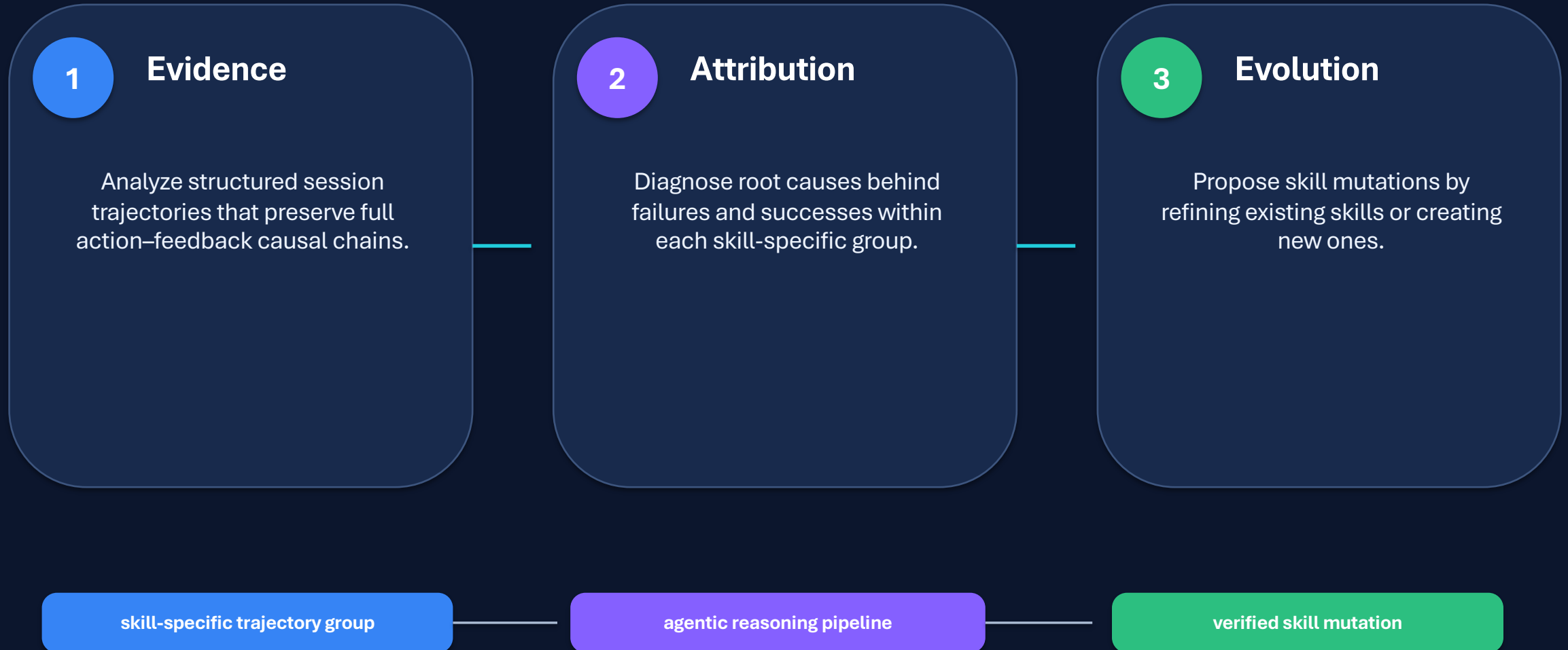


Architecture overview figure from source brief



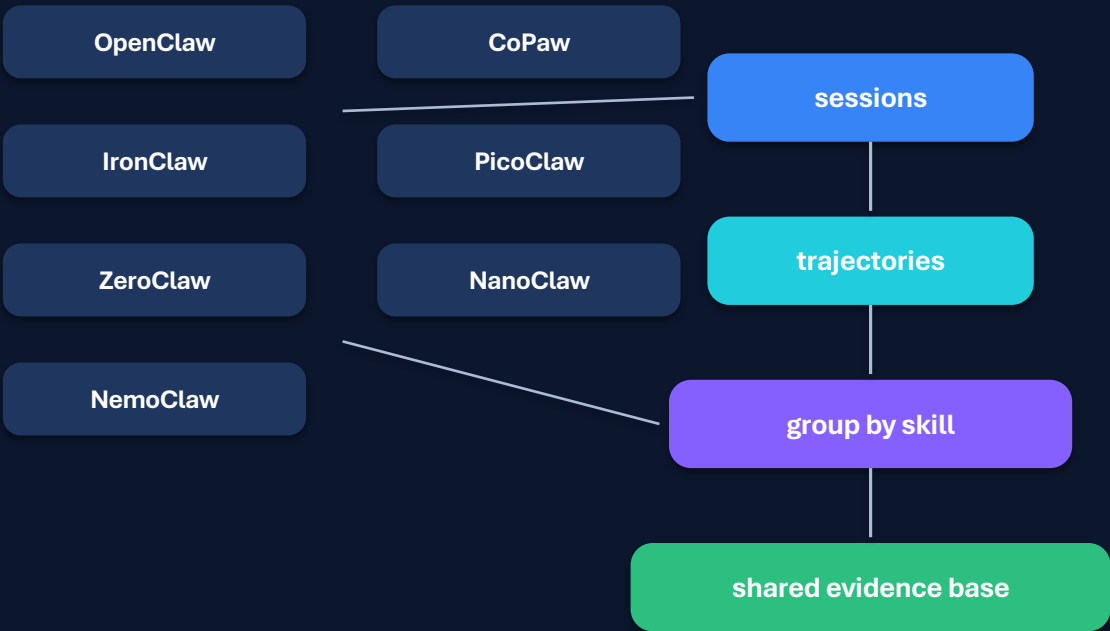
Agentic Evolver: Evidence → Attribution → Evolution

Open-ended autonomous reasoning over interaction evidence, not predefined update rules



Cross-User Trajectory Aggregation Exposes Behavioral Boundaries

Diverse contexts reveal where a skill works, breaks, and generalizes



Why aggregation matters

Different users exercising the same skill under diverse contexts produce complementary views of its behavioral boundary — conditions where it works and conditions where it breaks. A single user rarely provides enough signal to separate a generalizable improvement from an idiosyncratic fix.



complementary signals define the boundary

Case Study: Multimodal Creative Pipeline Skill

Six days of candidate evolution attempts; conservative verification prevents regressions

Day	Candidate skill	Brief rationale / skill function	Outcome	Next-day action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & environment setup	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Multimodal creative pipeline: extract from PDF/video/image	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual generation, structured validation	Reject	Keep current best pool
5	pipeline (impr.) + validate-required-input-files	Creative pipeline + per-file fail-fast validation	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / document-to-visual paths	Reject	Not admitted to next cycle

1 of 6 accepted

Conservative validator admits only improvements that beat the current best pool.

Case Study: Git Push Auth-Fallback Skill

Six-day log shows iterative refinement followed by plateau

Day	Candidate skill	Brief rationale / change summary	Update outcome	Next-day action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames; reduced inconsistency	Accept	Keep updated best pool
3	git-push + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	Accept	Keep current best pool
4	(none)	Same-pool retest; validator confirmed current pool as best	No update	Continue same best pool
5	git-push-with-auth-fallback	Added “push hang treated as auth failure”; no improvement	Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	Reject	Not admitted to next cycle

3 of 6 accepted updates

Evidence supports refinement for three days, then the current best pool plateaus.

Key Takeaways: What Distinguishes SkillClaw

Implications for continuously improving multi-user agent systems

Collective evolution

Individual interactions improve a shared, continuously improving skill ecosystem.

Fully automatic

Runtime interaction drives skill evolution without manual curation or explicit user intervention.

Agentic evolution paradigm

Open-ended reasoning over evidence produces skill updates rather than predefined rules.

Implication: agent ecosystems can accumulate operational knowledge across users, verify updates conservatively, and distribute improvements continuously.

Source brief notes evaluation on WildClawBench using Qwen3-Max demonstrates substantial improvements across tasks.